ML11 Charts that show what the numbers say

Week 4, Clean, chart, question | Session 11 of 24 | Machine Learning | New library: matplotlib, seaborn

Today's question. How do I show which item earns most, and how sales vary, in a picture?

Objective. By the end: you can draw a labelled bar chart, a histogram and a box plot, and write one finding and one limitation.

> **WHERE THIS IS USED**
> Weather forecasts, football stats and mobile money summaries all use charts so people can see a pattern fast. A good chart makes one clear point. What is the one point you want your chart to make?

Before this session: ML10 Cleaning without hiding evidence. If you cannot explain the last example, use its error card with the instructor first.

Words for today

Word	What it means	Think of it like this
bar chart	Bars compare a number across categories.	Stacks of coins, one stack per item.
histogram	Bars show how many values fall in each range.	Sorting pupils into groups by height.
box plot	A box shows the middle half of the values, with a line at the median.	A summary of a team's scores in one picture.
axis label	Words that name what an axis measures, with its unit.	The heading on a column in a register.
title	One line saying what the chart shows.	The heading of a notice.
finding	A claim a chart supports, with a number.	Book earns the most money: UGX 297000.

Block A. A bar chart of revenue

Chart the revenue of each item from the practice shop. Label the axis with its unit.

Predict. Predict which bar will be tallest and roughly how tall. Write your guess on paper first.

Your guess, written before you run it

```
import matplotlib.pyplot as plt
import pandas as pd
shop = pd.read_csv("camp_shop.csv")
shop["revenue_ugx"] = shop["units"] * shop["unit_price_ugx"]
totals = shop.groupby("item")["revenue_ugx"].sum()
plt.bar(totals.index, totals.values)
plt.ylabel("Revenue (UGX)")
```

```
plt.title("Revenue by item in the practice shop")
plt.ylim(bottom=0)
plt.show()
```

Run it, then compare. Three bars appear. Book is tallest at 297000, then Juice at 150000 and Pen at 103000. plt is the chart tool from matplotlib. ylabel and title add words, and ylim(bottom=0) starts the axis at zero so bar heights are fair. One finding is: Book earns the most, UGX 297000 of 550000. One limitation is: the data is invented and has no costs.

Investigate.

1. Why must a bar chart start at zero?
2. Does the chart tell us why Book earns most?

Modify. Chart total units instead of revenue.

Block B. A histogram and a box plot

seaborn draws statistical charts and builds on matplotlib. Draw how units vary and how they differ by item.

Predict. Predict which item has the lowest median units. Write your guess on paper first.

Your guess, written before you run it

```
import seaborn as sns
sns.histplot(data=shop, x="units", discrete=True)
plt.title("Units per sale")
plt.show()
sns.boxplot(data=shop, x="item", y="units")
plt.title("Units per sale for each item")
plt.show()
medians = shop.groupby("item")["units"].median()
```

Run it, then compare. The histogram counts sales for each number of units and shows how they spread from 1 to 8. The box plot shows one box per item with a line at its median. Juice has the lowest median, 3.5 units, while Book has 5.0 and Pen 5.5. seaborn plots read straight from the table, which saves writing the grouping yourself.

Investigate.

1. What does the line inside a box mean?
2. Which chart shows how many sales had exactly 3 units?

Modify. Change the histogram so it uses unit_price_ugx instead of units.

Make

Make the same bar chart for the values 6, 9 and 4 for Book, Pen and Juice. Write one finding with a number and one limitation. Say which chart type you would use for daily rainfall, and why.

Save your work as ML11. Keep the worked example and put your own attempt below it. Write two sentences: what you changed and how you checked it.

Error card

You see	It means	Fix
NameError: name 'sns' is not defined	seaborn was not imported in this session.	Run import seaborn as sns first.
The chart does not appear	The cell that draws it has not run, or plt.show() is missing.	Run the import cell, then run the drawing cell again.

Exit ticket

A chart needs labels with units, an axis that starts at zero for bars, and one finding with a limitation.

AI use follows AI_USE_POLICY.pdf. In weeks 1 to 6 no AI-generated solutions are used for exercises, labs or build tasks. Practise for 20 to 30 minutes between classes by repeating the Make task with one value changed. If you are stuck for 30 minutes, save the error and bring it to class. A file saved on your laptop has not been shared until you use the route in START_HERE.pdf.